I am eager to apply for the role of research assistant, with a focus on neuronal resilience in Alzheimer’s Disease. My prior project, which studied about neurodegenerative disease model, sparking my passion for exploring the biological foundations of neuroscience mechanisms. I worked with induced pluripotent stem cells (iPSC)-derived cerebral organoids from both human and mouse cell lines to investigate the neural dynamics of these 3D structures. These organoids were cultured in vitro and sectioned at various developmental stages using a cryostat for morphological analysis, identifying neural progenitor cells, astrocytes, mature neurons, and region-specific developments. Additionally, with calcium imaging to track neural activity using a two-photon microscope. Through this project, I gained hands-on experience in wet lab techniques.

My academic background has provided me with a solid foundation in neuroscience, with introduction from molecular mechanisms to circuits and behavior levels. My coursework in computational neuroscience has further enhance my understanding by allowing me to simulate neuronal models and apply theoretical knowledge practically.

Moreover, I possess laboratory experience in electronics, physiological signal recording, and medical equipment calibration. This background has heightened my awareness of potential laboratory hazards and reinforced my commitment to maintaining a safe working environment. My previous role as a research assistant honed my decision-making and communication skills, as the project required a high degree of self-motivation and progress management. It involved adaption and modification of experimental procedures to achieve optimal results. Completing a report at the end of this project enhanced my abilities in literature research, collaboration, and project management.

I believe my academic and practical experiences align well with the requirements of this position. I am particularly enthusiastic about furthering my research skills and knowledge in neuroscience, with the goal of pursuing a PhD in this field. My interest in neuroscience was initiated during my undergraduate project, which aimed at extracting features from EMG signals to improve muscle control via auditory feedback. Although this project finished up with a prototype successfully mapping EMG to acoustic signals, it underscored the need for better interpretation biological mechanisms to enhance the technology for therapeutic solutions. Throughout my academic journey, I have developed proficiency in MATLAB, Python, real-time data programming (with Arduino) and neural data decoding using machine learning techniques. I am committed to overcoming research challenges and understand that this process requires continuous learning and dedication.

I am excited about the opportunity to contribute to your research team and develop treatments for Alzheimer’s disease. Thank you for considering my application. I look forward to the possibility of discussing how my background, skills and enthusiasm could align with the goals of your research group.